# STACKS, DISCRIMINANTS: LEMMA-COMPARE-DUALIZING

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 3359-3375 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 3377-3431, the complete author proof (53 body lines).

*Editorial note.* Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 5 of queue_discriminant.json (pb=2373, stmt 3359-3375, proof 3377-3431).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0AUE: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0AX2: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0AA0: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0A9E: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0A9Y: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BT2: sources/stacks-discriminant/pinned-original-source.tex, source lines 380–387. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 05K0: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01SB: sources/stacks-discriminant/pinned-original-source.tex, source lines 3431–3431. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0BVG: sources/stacks-discriminant/pinned-original-source.tex, source lines 649–660. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : Y \to X$ be a quasi-finite separated morphism of Noetherian schemes. For every pair of affine opens $\mathrm{Spec}(B) = V \subset Y$, $\mathrm{Spec}(A) = U \subset X$ with $f(V) \subset U$ there is an isomorphism*

$$H^0(V, f^!\mathcal{O}_X) = \omega_{B/A}$$

*where $f^!$ is as in Duality for Schemes, Section 0A9Y. These isomorphisms are compatible with restriction maps and define a canonical isomorphism $H^0(f^!\mathcal{O}_X) = \omega_{Y/X}$ with $\omega_{Y/X}$ as in Remark 0BVG. Similarly, if $f : Y \to X$ is a quasi-finite morphism of schemes of finite type over a Noetherian base $S$ endowed with a dualizing complex $\omega_S^\bullet$, then $H^0(f^!_{new}\mathcal{O}_X) = \omega_{Y/X}$.*

*Proof.* By Zariski's main theorem we can choose a factorization $f = f' \circ j$ where $j : Y \to Y'$ is an open immersion and $f' : Y' \to X$ is a finite morphism, see More on Morphisms, Lemma 05K0. By our construction in Duality for Schemes, Lemma 0AA0 we have $f^! = j^* \circ a'$ where $a' : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_{Y'})$ is the right adjoint to $Rf'_*$ of Duality for Schemes, Lemma 0A9E. By Duality for Schemes, Lemma 0AX2 we see that $\Phi(a'(\mathcal{O}_X)) = R\mathcal{H}om(f'_*\mathcal{O}_{Y'}, \mathcal{O}_X)$ in $D^+_{QCoh}(f'_*\mathcal{O}_{Y'})$. In particular $a'(\mathcal{O}_X)$ has vanishing cohomology sheaves in degrees $< 0$. The zeroth cohomology sheaf is determined by the isomorphism

$$f'_*H^0(a'(\mathcal{O}_X)) = \mathcal{H}om_{\mathcal{O}_X}(f'_*\mathcal{O}_{Y'}, \mathcal{O}_X)$$

as $f'_*\mathcal{O}_{Y'}$-modules via the equivalence of Morphisms, Lemma 01SB. Writing $(f')^{-1}U = V' = \mathrm{Spec}(B')$, we obtain

$$H^0(V', a'(\mathcal{O}_X)) = \mathrm{Hom}_A(B', A).$$

As the zeroth cohomology sheaf of $a'(\mathcal{O}_X)$ is a quasi-coherent module we find that the restriction to $V$ is given by $\omega_{B/A} = \mathrm{Hom}_A(B', A) \otimes_{B'} B$ as desired.

The statement about restriction maps signifies that the restriction mappings of the quasi-coherent $\mathcal{O}_{Y'}$-module $H^0(a'(\mathcal{O}_X))$ for opens in $Y'$ agrees with the maps defined in Lemma 0BT2 for the modules $\omega_{B/A}$ via the isomorphisms given above. This is clear.

Let $f : Y \to X$ be a quasi-finite morphism of schemes of finite type over a Noetherian base $S$ endowed with a dualizing complex $\omega_S^\bullet$. Consider opens $V \subset Y$ and $U \subset X$ with $f(V) \subset U$ and $V$ and $U$ separated over $S$. Denote $f|_V : V \to U$ the restriction of $f$. By the discussion above and Duality for Schemes, Lemma 0AUE there are canonical isomorphisms

$$H^0(f^!_{new}\mathcal{O}_X)|_V = H^0((f|_V)^!\mathcal{O}_U) = \omega_{V/U} = \omega_{Y/X}|_V$$

We omit the verification that these isomorphisms glue to a global isomorphism $H^0(f^!_{new}\mathcal{O}_X) \to \omega_{Y/X}$. □